

QMS SOP TEMPLATE:

Deviation Management

Quality
Management
System
Standard
Operating
Procedure



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Introduction

In the highly regulated pharmaceutical industry, it is critical to ensure that any unanticipated changes, discrepancies, or failures in processes, systems, or materials are promptly addressed to safeguard product quality, patient safety, and regulatory compliance.

Unexpected events, excursions and discrepancies from previously established acceptable parameters and attributes may be detected at any stage of the process or during quality control activities which may include; the qualification, approval or release of facilities, cleaning, utilities, raw materials, packaging, process, executed procedures, testing, drug substance, and drug product.

These discrepancies may be minor and have no impact on the quality of the process or product. However, some events may constitute deviations that compromise the quality of the process or product, potentially leading to the rejection or recall of the product. To comply with regulation and to ensure that patients receive product that meets the established safety, purity, strength and quality, these events need to be investigated, managed and documented. This procedure is intended to guide the deviation management process to ensure that deviations are properly recorded, investigated, and resolved in a timely and compliant manner.

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1.0 Purpose

The purpose of this procedure is to describe the methodology behind the Deviation Management Quality System and link this process to other processes such as Root Cause Analysis (RCA) and the CAPA process, ensuring that deviations are properly recorded, investigated, and resolved in a timely and compliant manner.

Very specific deviations, like out-of-specification results, out-of-trend results and Environmental Monitoring deviations are discussed in more detail in their respective SOPs.

2.0 Scope

This procedure applies to all events and deviations that occur during manufacturing, analytical testing, release, transfer of material or product, storage, and distribution of a GMP product as well as Critical System Failures, and Facilities and Utilities Events.

This departure or discrepancy can be from a:

- Defined GMP process/procedure/protocol/instruction
- Defined GMP qualification/validated status
- Defined/specified GMP result (e.g., yield, specification)

3.0 Definitions/Abbreviations

Business Day: A business day is considered every official working day as defined by the business.

Contributing Cause: A cause that has been determined not to be the root cause of the incident/event but is found to be a contributory factor based on the investigation outcome and that must be considered when defining Corrective and Preventive Actions.

Corrective and Preventive Actions (CAPA): Corrective Actions are actions required to remediate an identified problem, investigate the discrepancy or non-conformance, analyze the information to identify the root cause and thus take corrective action to prevent recurrence. Preventive Actions are required to prevent occurrence of a similar potential discrepancy.

Deviation: An unplanned departure or unexplained divergence from an approved procedure, the applicable regulation, specification's acceptance criteria, tolerance limits, process control limits or written instructions that may affect product quality, safety, or compliance. Deviations are classified as Critical, Major or Minor as follows:

- **Critical deviation** is one that involves any of the following:
 - A variation from a defined critical process or standard that could or does result in product that does not meet its regulatory filed specification and/or required quality standard in each case when that product is no longer with the manufacturing site's control.
 - A significant quality incident that could, if not resolved, result in the failure to supply product which could impact patient treatment.
 - A significant Quality System failure that could go undetected and/or a significant purposeful withholding of information that may constitute fraud or falsification of products or data or give rise to the potential to harm the end user.

- **Major deviation** is one that involves any of the following:
 - Significant potential for product that has been produced or may be produced which does not comply with its registered/required quality and/or specifications and remains in control of the Manufacturer.
 - A significant breach of GMP that could affect the quality of the product but remains in control of the Manufacturer and does not fall under the definition of “critical”.
 - A collection of similar identified “minor” deviations that could indicate a specific control system failure.
- **Minor deviation** is one that involves any of the following:
 - A deviation that is not classifiable as “critical” or “major”, but indicates a departure from GMP.
 - By themselves, this type of deviations may not necessarily indicate a system failure, or would not have an impact upon the quality of the product, but represents a departure from GMP.

Effectiveness Check: The action to verify or validate the corrective and preventive action to ensure that it produced the result intended and that it does not adversely affect the product.

Impact Assessment: A determination of the impact the deviation has to product safety, quality, identity, strength, and/or purity, including impact on product stability, documentation and potentially related products, processes and/or systems.

Immediate Actions: A short-term action taken to quickly contain or resolve the issue or a progression of process steps with specific/additional checks and measures.

Risk Assessment: A systematic process of organizing information to support the risk management process. It consists of the identification of hazards and the analysis and evaluation of risks associated with exposure to those hazards.

Root Cause Analysis (RCA): A process employing problem-solving techniques used to determine root cause. The root cause analysis applied must be commensurate with the levels of risk associated with the deviation.

4.0 Responsibilities

Initiator: Responsible for initiation of the record within the next business day of discovery of the event. Attach all the supporting documentation available during the discovery process and initiation of the investigation.

Assigned to/Deviation Owner: Has the experience, training and ultimate responsibility for handling the event/deviation. Collect relevant information and records as well as assemble the investigation team and request technical assessment/product impact assessments.

Deviation Coordinator: Manages the process, tracks timelines, organizes meetings on deviations, summarizes information. (Not all facilities have this role.)

Quality Assurance/Qualified Person: Final responsibility for quality assessment and oversight of the deviation process.

Note: The term Qualified Person is not used in the US regulation. The responsibilities of the QP as defined in the EU are included among various individuals required to be diligently qualified and who are part of the Quality Unit. Each company is responsible for designing their Quality Management System.

Supervisors/Managers: Assigns, reviews, and approves deviations and ensures corrective and preventive actions are aligned with cGMP requirements, this SOP and company policies. Ensure personnel records and training are up to date. Communicate events and QA assessments between shifts to ensure all instructions are captured and followed.

Quality Assurance: Within Quality, it is the area responsible to respond to the incident/event with the area leader and support the impact and risk assessment process as well as the investigation process. Ensure the investigation is complete, Good Manufacturing Practice (GMP)/Standard Operating Procedure (SOP) compliant and that the root cause, impact assessment, CAPA identification and disposition are appropriate and complete. (Not all facilities have this role.)

5.0 Requirements

- Deviation forms
- Deviation log

6.0 Risks/Points of Attention

Ensure sure that immediate actions are taken to mitigate further impact from any possible Deviation.

Communicate any possible deviation to the Deviation Coordinator and Quality Assurance, within 24 hours from the date of discovery.

Special attention is paid to any suspicion of falsification/data integrity: in that case, the event is immediately escalated to the Quality Assurance/Qualified Person.

All deviations shall have a unique identification number, issued by the Deviation Coordinator or electronic system.

No deviations will be cancelled without justification, and no records related to Deviations will be discarded.

An appropriate cross-functional team should be assembled to review the situation, determine any immediate actions required and perform an initial impact assessment.

The cross-functional team should consist of the following individuals:

- Deviation discoverer
- Area Supervisor or Manager (or delegate)
- Quality Representative
- Subject Matter Experts (as necessary) e.g., Facilities, Maintenance, Supply Chain, Validation, Engineering, Product Development, etc.

Although date due will be established as 30 working days. Prioritization of investigations will be based on the criticality of the deviation.

7.0 Summary/Flowchart

- 7.1** Incident occurs, notify area supervisor immediately.
- 7.2** First assessment performed, together with Team Coordinator and Quality Assurance to determine if the incident is a deviation from an established procedure or process.
- 7.3** Possible Deviation is reported to the Deviation Coordinator, a number is issued and a form is filled out by the Team Coordinator or designee.
- 7.4** Form is returned to the Deviation Coordinator and assessed for completeness, Impact and Risk Assessment.
- 7.5** If the Risk Assessment is above the threshold, the Deviation Coordinator initiates a further investigation.
- 7.6** After the Investigation has been completed, the Deviation Coordinator closes the Deviation and proposes a final disposition.
- 7.7** The Deviation with, if applicable, the associated Investigation, is reviewed and approved before release of associated product by the Quality Assurance/Qualified Person.

8.0 Procedure

- 8.1** When an unplanned departure or unexplained divergence from applicable regulation, specification's acceptance limits, tolerance limits, process control limits or written instructions occurs (during manufacture, testing, receipt, storage, distribution, release) immediately contact the Team Coordinator/Supervisor of the area or the process.
- 8.2** If there are people at risk, or if there is any risk to the product or process, immediate action is taken, to contain or resolve the issue, if possible. This action can be e.g., discontinuing the process, switching to another workstation and/or segregation of material.
- 8.3** When a possible deviation occurs, this needs to be communicated to the Deviation Coordinator and Quality Assurance within 24 hours of discovery. A unique ID number needs to be requested, and a Deviation Form needs to be initiated by the Team Coordinator or designee.

Note: If there is any doubt about whether the incident/event constitutes a deviation, the Deviation Coordinator and Quality Assurance should be contacted. If it is determined, in collaboration with QA and the Deviation Coordinator, that the event is *not* a deviation, the incident/event should be recorded within the batch documentation, analysis sheet or other activity related documentation.

- 8.4** The Deviation Form is initiated by requesting a number and filling in the required details:

- Deviation Number
- Title/Short Description
- Who discovered the Incident/Event
- What happened and why is this considered a possible deviation?
- When was it discovered (Date and time)

- Where (the location where it happened)
- Description of Incident/Event and reference to governing documents, limits or ranges (e.g., SOPs, batch records, protocol)
- Any immediate action(s) taken? who was contacted? (QA, Engineering, Validation)
- Materials/Batch(es)/Areas or equipment possibly impacted
- Any attachments
- Date of initiation and name of initiator of the Deviation. If not initiated within the 24 hours of discovery, an explanation should be provided within the record.

8.5 After the Deviation Form is received by the Deviation Coordinator:

- A first review of details as submitted by the form, for clarity, completeness and correctness.
- A preliminary risk assessment related to the process and/or product is performed in collaboration with the cross-functional team.
- Should the outcome of the risk assessment reach a threshold, a further investigation is performed into the root cause(s) of the Deviation.

8.6 A cross-functional team with members of the departments involved, including the Deviation Coordinator will assess the Deviation.

8.7 After the Investigation is performed, the Deviation Coordinator will summarize the Investigation, including root cause(s), Batches impacted, a Final Impact Assessment and a Final Risk Assessment. This information, including the conclusion is entered on the second page of the form, for review by the Quality Assurance/Qualified Person.

8.8 All Deviations, including the appropriate short-term actions taken, together with a further Investigation into the root cause of the Deviation, and additional long-term actions (as CAPAs) proposed, need to be reviewed by the Quality Assurance/Qualified Person within 30 working days after initiation, prior to release of the product (if associated).

8.9 Deviations should be trended on root causes and any indication of specific or recurring problems requiring attention. This should be part of both an Annual Product Review (when it is considered product related) and regular Management Reviews.

9.0 Calculations

Calculation of Risk for Impact Assessment (see Addendum I)

10.0 Documentation of Results

- Deviations are recorded on a Deviation Form
- A reference to all deviations that are associated to a batch is listed on the release papers.
- A log with all deviations that are issued is maintained.

11.0 References

- 21 CFR Part 210 – Current Good Manufacturing Practice in Manufacturing, Processing, Packing, or Holding of Drugs; General
- 21 CFR Part 211 – Current Good Manufacturing Practice for Finished Pharmaceuticals
 - 21 CFR Part 211.100 (a) – Written procedures, deviations
 - 21 CFR Part 211.192 – Production record review
- 21 CFR Part 820 – Quality System Regulation (for Medical Devices)
 - 21 CFR Part 820.100 – Corrective and Preventive Action
- ISO 9001:2015 – Quality Management Systems
- ISO 13485:2016 – Medical Devices Quality Management Systems
- FDA Guidance for Industry: Quality Systems Approach to Pharmaceutical cGMP Regulations (2006)
- ICH Q7A – Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients
- ICH Q10 – Pharmaceutical Quality System
- EU GMP Guide – Part I: Chapter 1 (Pharmaceutical Quality System)
- EU GMP Guide – Part I: Chapter 5 (Production)
- EU GMP Annex 15: Qualification and Validation, Section 1.9
- Pharmaceutical Inspection Co-operation Scheme. *PIC/S Guide to Good Manufacturing Practice for Medicinal Products: Part I (PE 009-17)*
- TRS 986 – Annex 2: WHO Good Manufacturing Practices for Pharmaceutical Products: Main Principles
- Deviation Reporting for Human Cells, Tissues, and Cellular and Tissue-Based Products Regulated Solely Under Section 361 of the Public Health Service Act and 21 CFR Part 1271, Guidance for Industry

12.0 Revision History

- Rev 01 – Initial release
- Rev 02 – [To be updated with future changes]

13.0 Addenda

13.1 Addendum I: Preliminary Impact and Risk Assessment Example

Impact (Severity – what are the consequences of the deviation):

- Product/Process at risk to be rejected/unsuitable for use: 5 (severe)
- Some product or process impact, but mitigatable: 3 (medium)
- No process or product impact: 1 (none)

Detection – what is the likelihood that the deviation is detected:

- Continuous monitoring is applied (e.g., electronic systems or multiple quality reviews): 1
- An issue that affects the entire batch, process, equipment and for which a test exists that allows for automatic detection: 1
- Routine visual checks performed during processing, testing and handling: 3
- Sporadic defects detectable during in-process checks: 3
- Event for which no means of detection are routinely applied: 5

Occurrence – how often does this deviation occur:

- Similar deviation within last week: 5
- Similar deviation within last month: 4
- Similar deviation within last quarter: 3
- Similar deviation within last year: 2
- No previous occurrence or less than once a year: 1

Impact × Detection × Occurrence = Risk

Further Investigation required? Risk ≥ 3

13.2 Addendum II: Deviation Form Example (Page 1 of 2)

Deviation Number: 12345	Title: <i>Insects detected in Packaging Suite 33</i>	
Date and time of occurrence: <i>15 NOV 2024, 11:00</i>	Noticed by: <i>ASBU</i>	Location: <i>Packaging Suite 33</i>
Date Deviation initiated: <i>15 NOV 2024</i>	By: <i>ASBU</i>	Department: <i>Packaging</i>
Description of Deviation in relationship to normal situation:	<i>During the line clearance for batch 77888, ISPEmab 10 mg/ml USA, I noticed a spider in the box of cartons with lot # 22322.</i>	
Why did the Deviation happen?	<i>Not sure if the spider was already in the box or if the spider crawled in after opening the box by production. Production team also had no idea.</i>	
Immediate Action Taken? Yes / No If Yes, describe.	<i>Spider was removed, packaging activities were stopped</i>	
Product/Material related: Yes / No	Product Name: <i>Cartons ISPEmab 10 mg/ml USA</i>	Batch Number(s): <i>77888</i>
Attachments (Yes / No)	List attachments (only if Yes): <i>Inspection findings of other boxes</i>	
Initial Impact Assessment (I): (If No impact, explain)	<i>Product cannot be packed using this material. High (5)</i>	
Risk Assessment	Occurrence (O): 3 Reference: <i>N/A</i>	Risk: $I \times O = 15$
Investigation Number, if Risk ≥ 3	<i>I – 24335</i>	Date Initiated: <i>15 NOV 2024</i>
Deviation Coordinator	Date: <i>15 NOV 2024</i>	By: <i>JEHO</i>

13.2 Addendum II: Deviation Form Example (Page 2 of 2)

Deviation Number: 12345	Title: <i>Insects detected in Packaging Suite 33</i>	
Investigation Closed:	Date: 26 NOV 2024	By: JOHO
Batches involved:	<i>All materials related to Supplier PackyTack – see attached ERP List</i>	
Conclusions of Investigations:	<i>Materials are contaminated with insects originating from Supplier PackyTack, as we have noticed in several other packages (see attached list)</i>	
Root Causes Established	<i>Insufficient Pest Control at PackyTack</i>	
Final Impact and Risk Assessment:	Impact: <i>High (5)</i>	Risk: <i>High</i>
CAPAs initiated?	<i>CAPA 24 555</i>	
Comments:	<i>N/A</i>	
Deviation Coordinator	Date: 26 NOV 2024	By: JEHO
Batch Disposition	<i>Packaging materials from Supplier PackyTack are to be returned for investigation. This was the first batch of PackyTack material used after 5 months, no other occurrence of insects has been noted. No complaints on insects received, no further action is taken.</i>	
Comments:	<i>The supplier is notified, an audit is requested. The materials will be bought from another approved supplier until further investigation has been performed.</i>	
Quality Assurance/Qualified Person for Closure		
Name: <i>J. Menno</i>	Date: <i>01 DEC 2024</i>	Signature: <i>JMenno</i>



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